



Pinnacle Points

Points from which no higher
point can be seen

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About Me: Jamie Breault

- Master of Data Science at UBC
- 6 years developing insurance software
- Undergrad in Physics at the University of Waterloo
- Avid backpacker
 - 52 multi-day hikes
 - Often wonder if I am at the highest point in sight



Data Sources

1. Global Summits

- 25 million summits with over 100 feet of prominence

2. OTOTW Mountains

- 6500 summits where no land rises above its horizontal plane

3. Open-Meteo's Elevation API

- Fetches elevation by latitude and longitude

4. Beyond Horizons

- 73 extreme lines-of-sight confirmed by photograph

Finding Pinnacle Points

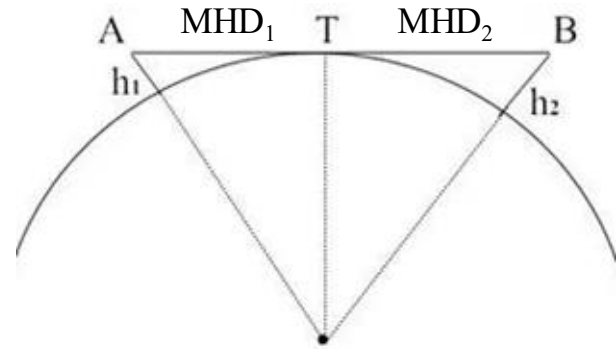
Every OTOTW mountain is a pinnacle point candidate. For each:

1. Validate higher local summits to test if line-of-sight analysis is worthwhile
2. Do line-of-sight analysis for summits that pass validation
 - Earth's curvature with topography
 - Atmospheric refraction
 - Atmospheric scattering
3. If no higher summits have line-of-sight with the candidate, then it is a pinnacle point

Maximum Horizon Distance (MHD)

- MHD is the maximum distance that can be seen from a point assuming all surroundings are at sea level
- Line-of-sight is impossible if the distance between the observer and target is greater than the sum of their MHDs

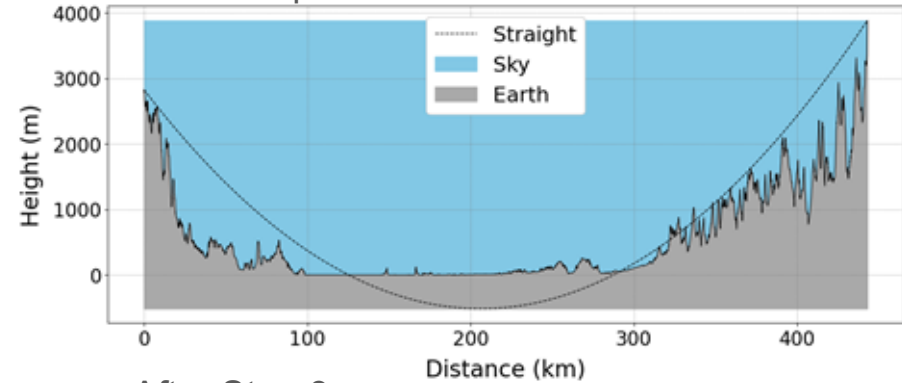
$$MHD = \sqrt{\frac{k}{k-1} 2R_{\oplus} h}$$



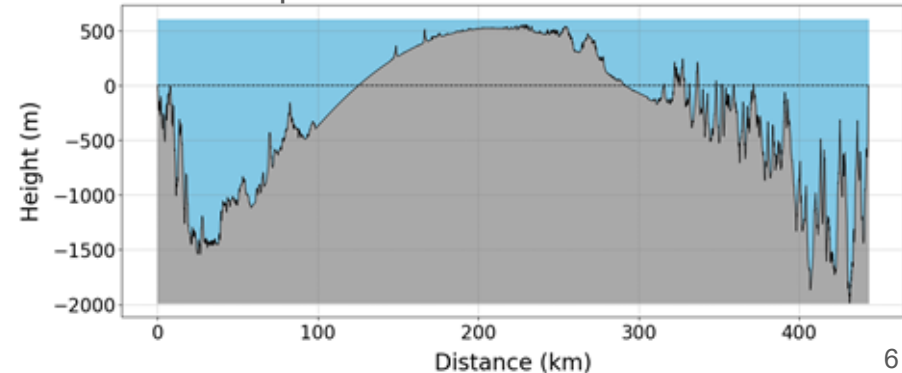
Earth's Curvature with Topography

1. Space points 100 m apart between the observer and target
2. For each, get elevation and distance from observer
3. Transform to a physically representative coordinate system

Before Step 3:



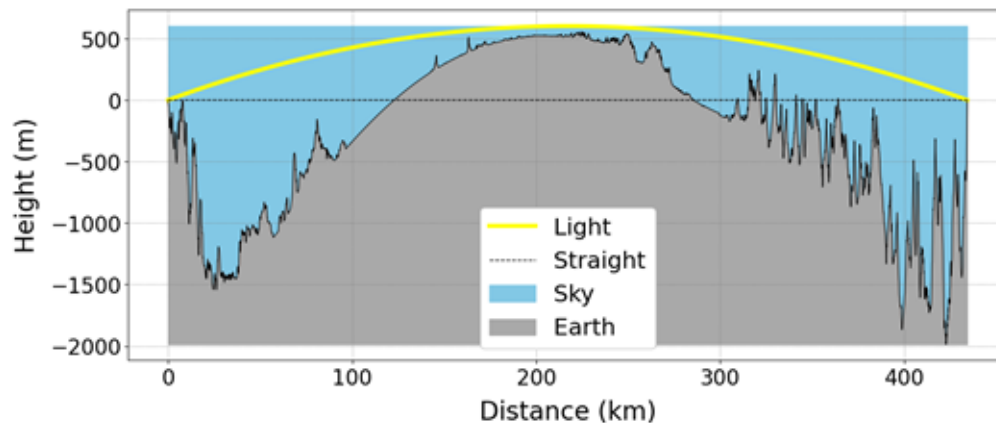
After Step 3:



Atmospheric Refraction

1. Determine the minimum light curvature needed for all confirmed lines-of-sight to be theoretically possible ($k=6.4$)
2. Use this curvature whenever doing line-of-sight analysis for pinnacle point candidates

$$R_{Light} = kR_{\oplus}$$



Atmospheric Scattering

1. A point's scattering coefficient (β) can be calculated from its altitude (h_L)

$$\beta = \beta_0 e^{-h_L/H}$$

1. Integrate over the light's path to get the average (β_{avg})

$$\beta_{avg} = \frac{1}{d_L} \int_0^{d_L} \beta \, dl$$

1. Calculate contrast (C), >0.02 is required for visibility

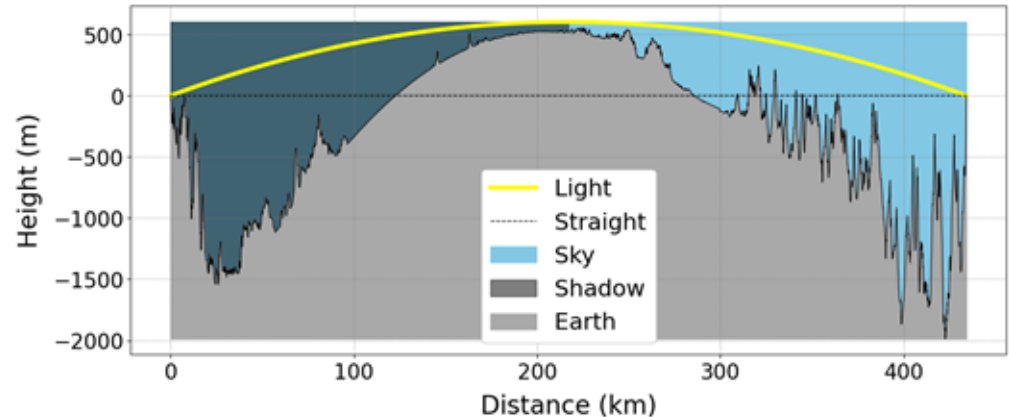
$$C = e^{-\beta_{avg} d_L}$$

Partial Irradiation

Extreme lines-of-sight are observed just before sunrise when there is partial irradiation to increase contrast.

Suppose there are two regions, the final region is fully irradiated and the first region is reduced by a factor α .

$$C = \frac{e^{-\beta_2 d_2}}{1 - \alpha + \frac{\alpha}{e^{-\beta_1 d_1}}}$$



Result and Validation

1554 pinnacle points were identified. An interactive map was shared online:

1. prominence@groups.io on Email
 - Mountain metric specialists
 - World-record-holding mountaineers
2. r/mountaineering on Reddit
 - Mountaineering enthusiasts

The 4 confirmed errors discovered so far are the result of inaccuracies in elevation data.



Next Steps

- Investigate alternate light modelling solutions based on feedback

Questions?

